AI-Powered Job Matching using Retrieval-Augmented Generation (RAG)

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1. Problem Statement

Most existing job portals rely on keyword-based matching systems, which often fail to capture the actual meaning of a candidate's skills and experience. This leads to irrelevant job recommendations and poor personalization.

For example, a candidate with experience in "BERT" or "transformers" may not be matched with roles requiring "Natural Language Processing," even though they are closely related. Similarly, basic keyword overlap can result in mismatched recommendations across different experience levels.

This highlights the need for a system that understands job descriptions and resumes **semantically rather than lexically**.

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2. Proposed Solution

I propose an **AI-powered job matching system using Retrieval-Augmented Generation (RAG)**.

The system combines:

- **Semantic retrieval** using embeddings
- **Reasoning capability** of Large Language Models (LLMs)

Instead of simple keyword matching, the system:

1. Retrieves relevant job postings using vector similarity
2. Uses an LLM to refine and rank results based on context

This enables more accurate and personalized job recommendations.

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3. System Design

The system is designed as a scalable full-stack architecture:

- **Frontend:** React (user profile, job search interface)
- **Backend:** Python (FastAPI) for ML integration
- **Embedding Model:** Sentence Transformers
- **Vector Database:** FAISS
- **LLM Layer:** OpenAI / open-source LLM

Pipeline:

1. Convert resumes and job descriptions into embeddings
2. Store them in a vector database

3. Retrieve top-k similar results
4. Use LLM to improve ranking and generate explanations

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4. Key Features

- Semantic job matching
- Personalized recommendations
- Resume understanding beyond keywords
- Context-aware ranking using LLMs

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5. Expected Outcome

- More relevant job recommendations
- Reduced mismatch between candidates and roles
- Improved user experience
- Scalable AI-driven job platform

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6. Future Work

- Resume parsing from PDFs
- Feedback-based learning system
- Domain-specific fine-tuning of embedding models